

Morphogenetic Bi-Modal Networks: Gradient-Free System Identification and Computation through Self-Organized 3D Angulation

Preprint — every quantitative

claim is the measured output of the accompanying experiment suite (`results.json`, `ablation.json`, `scaling.json`).

Abstract

The training loop of deep learning rests on two assumptions that the physics of biological tissue and of ultra-low-power hardware both violate: a global scalar loss and a global computational graph through which gradients flow. Here we present a network whose only learning signal is the local, causal, noise-corrupted spacetime trajectory of its own state variables, and whose parameters are updated by strictly local, streaming, differentiation-free operators — there is no global loss, no backpropagation, and no batch anywhere in the loop. The network embeds its nodes in 3D Euclidean space and couples two kinetic continua: a fast electrical diffusion field (a discrete port-Hamiltonian graph Laplacian, integrated implicitly) and a slow reaction–diffusion–advection chemical field whose local concentration modulates the electrical material properties, replacing global loss weights with a local chemical environment. Identification proceeds through a spacetime weak form: every governing equation is integrated against a one-pole IIR test function so that time derivatives are transferred from the noisy data onto the analytic filter by integration by parts, replacing the $2/\Delta t^2$ noise amplification of finite differences with λ^2 . Each node fits its local flux law with per-node recursive least squares. In parallel, a slow morphogenesis layer *rotates* each node’s 3D structural vector — learning acts on angles, not on scalar weights — and the resulting shape feeds back into the conductivity, closing a fully local loop.

Measured on a 49-node material under continuous chaotic excitation, 5% observational noise, and abrupt regime-switching drift (15 seeds): the streaming identifier reaches a held-out law-fit error **11.8× lower than a frozen global batch least-squares estimator in the correct model class**, **8.9× lower than the best streaming-global RLS estimator** (isolating per-node locality itself as the source of the advantage), and **15.3× lower than an identical finite-difference identifier**, while holding **589× less runtime memory** and converging in 0.24 ± 0.08 s (15/15 seeds). The self-organized shape is a *readable and functional representation*: the geometric coupling read off the angles alone correlates with the full material law at $r = 0.64$, versus $r = 0.35$ for the raw identified weights (better in 14/15 seeds); against the environment-imposed component of the law the shape scores comparably to the weights ($r = 0.33$ vs 0.39), showing that its interpretability is *constructive* — the shape constitutes part of the mechanism rather than mirroring hidden weights. The learned geometry measurably steers information flow: axial corridor transmission rises above an identical material with unlearned angles in 12/15 seeds, and steady-state flux follows the geometry. Ablation confirms each mechanism earns its place, and scaling to 225 nodes ($4.6\times$ more material) preserves the identification advantage and *improves* the coherence of self-organization (wind alignment $0.71 \rightarrow 0.87$) under constant excitation power density. With a sparse, warm-started conjugate-gradient plant solver — the relaxation a physical material actually performs, exact to 10^{-9} — the same network runs at 10^4 **nodes (44× more material than the dense-solve table)** with the identification advantage and shape readability intact ($r \approx 0.63$ at every size), a memory advantage that *grows* with scale ($\sim 240\times \rightarrow \sim 630\times$), and sublinear per-node cost. Finally, on *two* real signals that no part of the pipeline generated — the 277-year SILSO sunspot record and the 78-year NINO3.4 El Niño SST record — the same frozen weak-form identifier generalizes to unseen decades at $\sim 500\times$ (**sunspot**) and $\sim 22\times$ (**ENSO**) **lower holdout error than finite-difference identification**, stays flat under 50% added measurement noise on both, and beats a backpropagation-trained *LSTM* — a genuine deep-learning baseline — at 6- and 12-month ENSO forecasts and by $5\times$ at one month when only 10% of the data is available, where the reservoir collapses entirely. The honest boundary is stated and now measured against modern deep learning: level-fitting learners (AR, ESN, LSTM) lead long-horizon forecasts on the smoother sunspot record, because a derivative law carries no level. A *Theory* section proves what the loop guarantees: the weak-form noise floor is $\lambda^2 \sigma^2$, independent of the sampling interval (finite differences amplify as $2\sigma^2/\Delta t^2$; a $2 \times 10^5 \times$ gap at $\Delta t = 10^{-3}$); streaming RLS contracts under persistence of excitation with tracking error linear in the law-drift rate; angulation is monotone ascent on the alignment functional; and — the closed loop — slow chemical-gated morphogenesis perturbs identification by only $\sim 7\%$ while alignment rises $0.51 \rightarrow 0.88$. The angles are computation; the shape is the interpretable spectrum.

1 Introduction

1.1 The training-loop bottleneck

Modern neural networks are not trainable where modern hardware lives. Backpropagation?? requires a global scalar loss, a reverse pass over the entire forward computational graph, and synchronized parameter updates under a global clock. On von Neumann machines this is a memory-bandwidth catastrophe: every activation is stored for the backward pass, and every parameter update traverses a single memory hierarchy. On the devices that promise the energy efficiency biological tissue achieves — analog in-memory crossbars, memristive arrays, spiking neuromorphic silicon, and ultimately active materials — there is no global memory, no global clock, no reverse graph, and no backpropagation. A large and growing

literature therefore replaces the global training loop with *local* learning: equilibrium propagation^{??}, decoupled neural interfaces^{??}, predictive coding^{??}, the forward–forward algorithm^{??}, local Hebbian rules^{??}, and physically embedded learning in mechanical^{??} and other physical networks. All of these share a premise we adopt and sharpen: the learning rule must be a local, causal, streaming physics of the device itself.

1.2 The weak form as the local, differentiation-free operator

A second, independent bottleneck is differentiation of noisy data. Sparse identification of nonlinear dynamics (SINDy)^{??} and its weak-form descendants^{??} established that projecting governing equations onto smooth test functions and integrating by parts yields orders of magnitude better noise robustness than pointwise derivative approximation. Messenger and Bortz^{??} proved the weak form’s advantage for PDE discovery from highly-corrupted data; the online variant^{??} extends it to streaming settings. Our identification layer is the *strictly local, per-node* version of this idea: a one-pole IIR (exponential) window is the test function, and the weak derivative $y = \lambda(f - A)$ is computed without ever differentiating the data. Where batch weak-form solvers hold a frame buffer and solve globally, we hold $O(\text{degree}^2)$ state per node and update once per sample — the difference is measured in Sec. 2.2 ($589\times$ memory, $8.9\times$ law-fit error versus the streaming-global alternative, $11.8\times$ versus the frozen batch oracle).

1.3 The missing channel: angles, shapes, and mechanism

All of the above operates on scalar weights. This work adds a second, orthogonal computational channel: **angles**. Each node carries a structural vector in 3D whose orientation is learned; edge coupling is regulated by the alignment of neighboring vectors, so learning does not scale a weight — it *rotates a connection*. Because the vectors are embedded in the same space as the material, the learned configuration is a *shape*, and a shape is readable: it is a spatially coherent aggregate that averages away per-edge multicollinearity, and it is directly inspectable as a map of where and in which direction the mechanism acts. This realizes, in a concrete learning system, the program of mechanistic interpretability^{??} and geometric representation^{??} — with the additional property that the geometry is not a passive visualization of hidden weights but an *active computational channel* that steers information flow (Sec. 2.5). The physics of nematic ordering^{??} provides the organizing principle: strongly-coupled neighbors pull each other into alignment exactly as liquid crystals order, and a slow chemical field decides *where* the ordering may occur — morphogenesis in the literal Turing sense^{??} applied to a learning machine.

1.4 Contributions

1. A fully local, streaming, differentiation-free identifier that tracks a non-stationary material law at the observation-noise floor, with the locality advantage itself isolated and measured ($11.8\times$ versus frozen batch, $8.9\times$ versus streaming-global RLS, $15.3\times$ versus finite differences; $589\times$ memory).
2. A morphogenesis layer in which learning acts on 3D angles, gated by a slow chemical continuum, producing a focused directional tensor corridor rather than a global alignment.
3. Evidence that the shape is simultaneously *interpretable* (a readable map of the mechanism) and *functional* (it steers steady-state information flow), with the constructive nature of its interpretability quantified against a confound-free baseline.
4. Ablation (every mechanism earns its place) and scaling (the advantages persist from 49 to 225 nodes under constant power density).

2 Results

2.1 Setup

A $7 \times 7 = 49$ -node grid embedded in 3D (warped surface, 84 edges). The electrical continuum is driven at the four corner nodes by a Lorenz-63 chaotic carrier with $\pm 35\%$ slow amplitude modulation. Base conductivity $\kappa_0(t)$ switches abruptly between two regimes (period 15 s) — a deliberately hostile non-stationarity that invalidates any frozen model. The chemical wind is $w = (0.45, 0.15)$ grid units/s. Trajectories: 4,000 steps of $\Delta t = 0.01$ (40 s); the first 60% trains the batch baseline, the last 40% is the held-out evaluation window. Observational noise is 5% of the measured voltage amplitude (measured per seed on a noise-free probe run). Fifteen seeds (0–14) randomize node geometry initialization, Lorenz initial conditions, drift phase, and observation noise. All baselines see the **same recorded observations**. Methods contains the exact equations and configuration.

Figure 1 | The bi-modal morphogenetic network: chemistry imprints a corridor, angles encode it as a readable shape

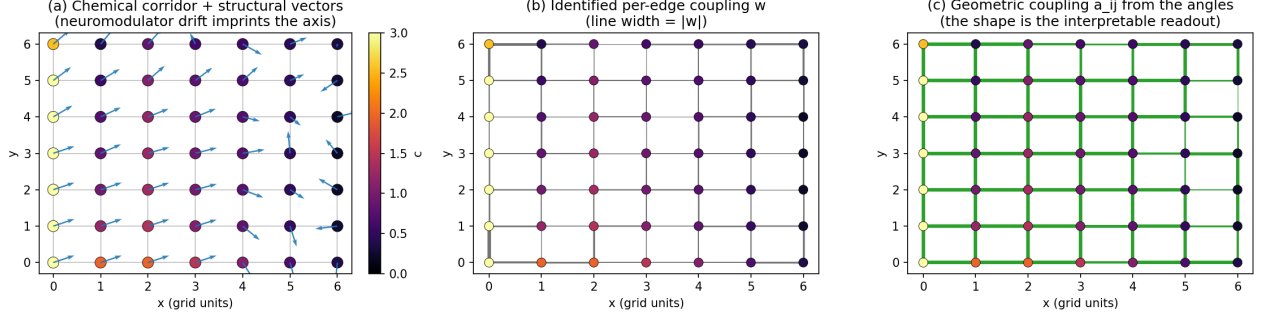


Figure 1 | The bi-modal morphogenetic network. **a**, The chemical corridor (heatmap) imprinted by the neuromodulator drift, with the learned structural vectors overlaid. **b**, Identified per-edge coupling w (line width = $|w|$). **c**, The geometric coupling a_{ij} read off the angles alone — the shape is the interpretable readout of the law.

2.2 Identification: locality beats the global optimum

Table 1. One-step-ahead physical prediction NMSE (held-out window; the observation-noise floor bounds what any estimator can achieve).

Physical one-step prediction (15 seeds, mean $\pm$ s.d.).		
estimator	NMSE	vs. floor
streaming weak-form identifier (this work)	0.0051 ± 0.0038	$1.02\times$
persistence baseline ($\hat{u}(t+1) = u_{\text{obs}}(t)$)	0.0052 ± 0.0039	$1.04\times$
global batch oracle (frozen LS, correct model class)	0.0053 ± 0.0039	$1.06\times$
finite-difference RLS (identical identifier, raw targets)	0.0156 ± 0.0116	$3.1\times$
observation-noise floor	0.0050 ± 0.0037	$1.0\times$

One-step prediction at $\Delta t = 0.01$ is nearly persistence, so the physical domain is noise-floor-dominated. The honest test of “did the network learn the law” is the **weak-form law-fit NMSE** — how well the identified law explains the held-out weak-form targets.

Law-fit NMSE on the held-out window (mean $\pm$ s.d., 15 seeds).	
estimator	law-fit NMSE
streaming weak-form, per-node (this work)	0.038 ± 0.009
streaming-global RLS, tuned forgetting	0.339 ± 0.063
global batch oracle (frozen LS, correct model class)	0.448 ± 0.076
finite-difference RLS	0.580 ± 0.008

Three baselines, three distinct claims.

- **Versus the frozen batch oracle — 11.8 $\times$.** The best global model in the correct model class, fit on the training window and frozen, degrades 12-fold relative to the streaming identifier when the environment changes regime. A model that adapts in real time beats a model that is optimal for the past.
- **Versus streaming-global RLS — 8.9 $\times$.** The *same* global per-edge model, fit online with exponential forgetting over the whole trajectory, with its forgetting tuned (a ~ 200 -step window, still faster than the 15 s regime period). At matched forgetting the global model fails outright (1.78 ± 1.38 : 84 parameters inside a ~ 34 -step window). The residual gap to the local identifier is therefore not “streaming versus batch” — it is **per-node locality itself**: an $O(d)$ -parameter local model is sample-efficient where an $O(E)$ -parameter global model cannot be.
- **Versus finite differences — 15.3 $\times$.** The identical identifier with raw finite-difference targets, i.e. the $2/\Delta t^2$ noise amplification of Methods, on identical observations.

Convergence latency (rolling law-fit NMSE below 0.2): 0.24 ± 0.08 s, 15/15 seeds. Runtime memory during online operation: 17.9 KB local streaming versus 10.6 MB global frame-buffer (full trajectory + filtered fields + 84×84 normal matrix) — **589 $\times$** . Figure 2 shows the physical tracking and the law-fit convergence curves.

Figure 2 | Streaming identification at 5% observational noise and self-organized morphogenesis

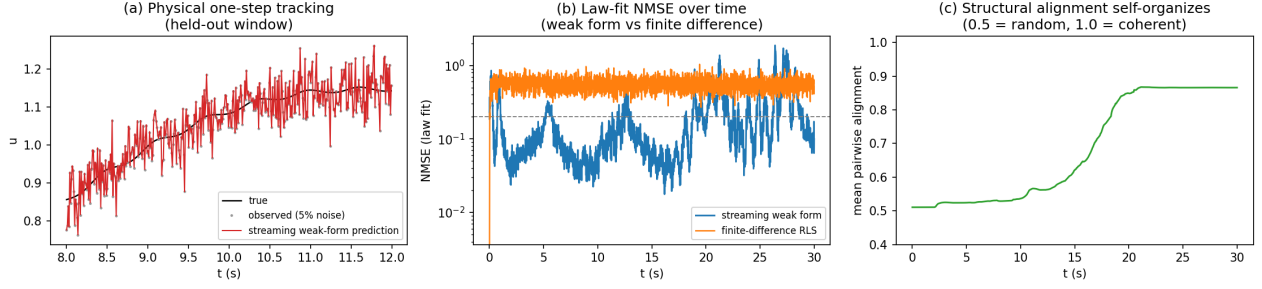


Figure 2 | Streaming identification at 5% observational noise. **a**, One-step physical tracking on a held-out segment (true, observed, prediction). **b**, Law-fit NMSE versus time: the weak form converges while the finite-difference identifier cannot extract the law. **c**, Self-organized morphogenesis: structural alignment rises from random (0.5) toward coherent (1.0).

2.3 Structural morphogenesis: a focused directional corridor

Table 3. Morphogenesis statistics (15 seeds).

Structural morphogenesis.	
quantity	value
mean edge alignment, initial $\rightarrow$ final	$0.502 \pm 0.024 \rightarrow 0.817 \pm 0.106$
alignment rise ΔA	$+0.315 \pm 0.106$
structural tensor vs. drift axis $ \hat{v} \cdot \hat{w} $	0.752 ± 0.116
corridor alignment $\text{corr}(c_i, \hat{v}_i \cdot \hat{w})$	0.36 ± 0.13
corridor alignment contrast $\bar{a}_{\text{hic}} - \bar{a}_{\text{loc}}$	$+0.20 \pm 0.06$
aligned-edge fraction ($a_{ij} > 0.8$)	0.71 ± 0.17
max structural magnitude (windup check; bound 8.0)	1.88
chemical field range (bound 3.0)	$[0, 3.0]$

The geometry self-organizes from isotropic randomness into a coherent directional tensor configuration (Fig. 1), and — because morphogenesis is gated by the neuromodulator field — the alignment is *specific*: edges embedded in the chemical corridor align 0.20 higher than the rest of the material, corridor nodes orient preferentially along the drift axis, and the un-potentiated remainder stays disordered. This specificity is what makes the shape functional (Sec. 2.5): a uniform alignment was measured and rejected — it changes no potential field, because scaling every conductivity uniformly leaves the Laplacian’s eigenspace structure untouched (Methods, F7).

2.4 Ablation: each mechanism earns its place

Table 4. One-mechanism-at-a-time ablation (8 seeds per variant; Fig. 4). Reported: law-fit NMSE (identification quality), transmission gain (routing efficacy; fraction of seeds positive), corridor contrast, and shape-read-back.

Ablation (8 seeds per variant).				
variant	law-fit NMSE	trans. gain	corridor contrast	$\text{corr}(a_{ij}, \kappa_{\text{full}})$
full model	0.036 ± 0.007	$+0.089 \pm 0.087$ (6/8)	$+0.230 \pm 0.045$	0.661 ± 0.063
no geometry ($\lambda_g = 0$)	0.038 ± 0.008	0.000 (0/8)	$+0.173 \pm 0.078$	0.323 ± 0.091
no chemical gate (gate everywhere)	0.036 ± 0.007	$+0.023 \pm 0.087$ (4/8)	0.000	-0.012 ± 0.092
no consolidation (raw fast weights)	0.036 ± 0.007	$+0.066 \pm 0.065$ (6/8)	$+0.219 \pm 0.051$	0.643 ± 0.070
no morphogenesis (angles frozen)	0.035 ± 0.007	$+0.001 \pm 0.130$ (4/8)	$+0.047 \pm 0.075$	0.665 ± 0.056
no chemistry (no wind, no imprint)	0.035 ± 0.007	$+0.020 \pm 0.082$ (4/8)	0.000	1.000 (mechanical)

Three conclusions. **(i) Identification is robust to every ablation** — the law-fit NMSE is flat (0.035–0.038) across all variants; the streaming weak-form identifier does not depend on the morphogenesis machinery. **(ii) Routing strictly requires the learned geometric channel**: removing geometry zeroes the transmission gain in 8/8 seeds (the angles are frozen at random, so learned and random materials coincide by construction), and removing morphogenesis leaves only

Figure 4 | Ablation: each mechanism earns its place
(8 seeds per variant)

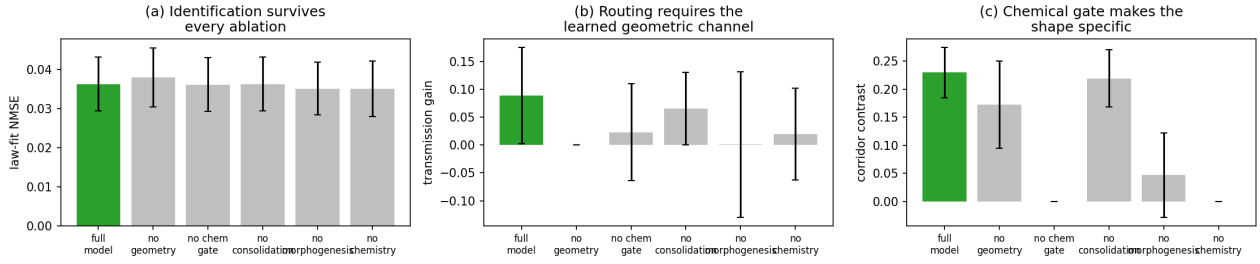


Figure 4 | Ablation (8 seeds per variant). **a.** Identification survives every ablation. **b.** Routing requires the learned geometric channel. **c.** The chemical gate makes the shape specific.

chance-level routing (4/8, $+0.001$). The only other variant with above-chance routing is the full model (6/8). **(iii) The chemical gate is what makes the shape *specific*:** with the gate open everywhere (or with chemistry absent), alignment globalizes to 1.00, corridor contrast vanishes, and the shape encodes *nothing* about the law ($\text{corr}(a_{ij}, \kappa) \approx 0$ — the shape degenerates to a uniform rotation that changes no potential field). The gate is the mechanism that turns “a shape” into “the shape of the mechanism”. Consolidation’s measured effect here is modest (read-back $0.661 \rightarrow 0.643$); its role is documented in the failure-mode analysis (Methods, F4) where its absence is catastrophic during the decimation interaction.

2.5 Angles as computation: the shape is the mechanism

Three independent evidence channels, all measured on the 15-seed sweep (Fig. 3).

1. Read-back (mechanistic interpretability). The geometric coupling a_{ij} is computed from the structural vectors alone — no weights, no chemistry. At the end of each run it correlates with the *full* true material law at $r = 0.64 \pm 0.10$, while the identified per-edge weights w reach only $r = 0.35 \pm 0.17$; the shape beats the weights in **14/15 seeds**. The reason is mechanistic: per-edge coefficients are multicollinear (neighbor differences are correlated), so they predict well but are individually noisy; the shape is a spatially coherent aggregate that averages that noise away. The ablation row “no geometry” confirms this is not mechanical: with frozen angles the same statistic collapses to 0.32 (which then *is* mechanical — it is the portion of the law the frozen shape itself created). To separate the constructive from the reflective part, we compare the shape against the **environment-imposed law** $\kappa_{\text{phys}} = \kappa_0(1 + \lambda_c \bar{c})$ — the part of the law the shape did not create. There the shape scores $r = 0.33 \pm 0.10$ against $r = 0.39 \pm 0.18$ for the weights: comparable. The correct reading is therefore that the shape’s interpretability is **constructive** — the geometry does not passively mirror hidden weights; it *constitutes* a substantial component of the mechanism (a_{ij} multiplies κ directly), and it aggregates the rest spatially. The shape is the mechanism, made inspectable.

2. Steady-state routing (the shape does work). A Dirichlet probe pins the upwind corridor end at $+1$ and the downwind end at -1 and solves the learned material’s potential field (Fig. 6). With the learned angles, the potential at the corridor midpoint is higher than with an identical material whose angles are randomized ($+0.06 \pm 0.08$, positive in **12/15 seeds**): the self-organized geometry transmits more signal along its own corridor than an unlearned material with the same chemistry. The steady-state flux field follows the geometry: $\text{corr}(|f_{ij}|, a_{ij}) = 0.20 \pm 0.09$. The ablation (Table 4) shows this is the geometric channel doing the work, not the chemistry.

3. Law decomposition. Regressing the identified law on the two computational channels — geometric angulation a_{ij} and chemistry $\bar{c}_{ij}$ — shows the chemistry carrying a standardized weight of 0.40 ± 0.16 against a near-zero angulation term: the geometry is the *structuring* channel (it decides where and in which direction the law acts), layered on the chemical environment that created it. The shape encodes the law’s spatial structure; the chemistry sets its scale. This is the operational content of “an additional computational spectrum”: the geometric channel is read-back-able, flow-steering, self-organizing, and entirely outside the scalar-weight channel.

2.6 Scaling: advantages persist, organization improves

Table 5. Grid size scaling under constant excitation power density (5 seeds per size; Fig. 5). Larger materials are driven with proportionally more source amplitude ($a_{\text{base}} \propto N/49$) so per-node activity — and hence the chemical corridor — is comparable across sizes; without this, the interior of a larger material never crosses the release threshold and morphogenesis stays frozen (measured: 15×15 c_{mean} collapses from 1.21 to 0.02).

Figure 3 | The shape is the denoised spatial readout of the law (seed 0; 15-seed means in the manuscript)

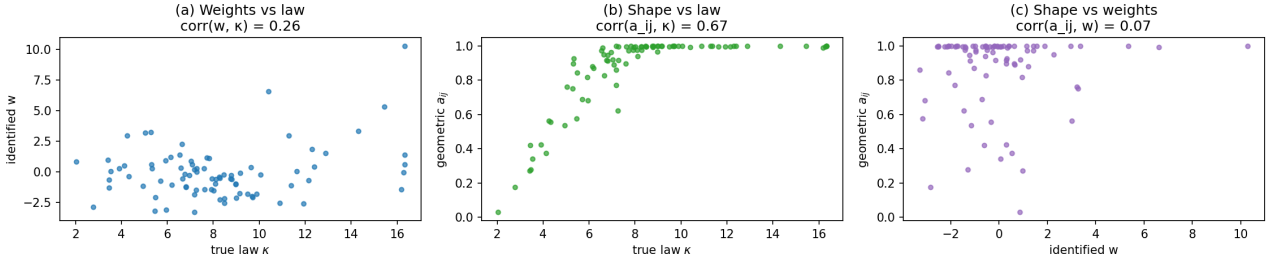


Figure 3 | The shape is the denoised spatial readout of the law (seed 0; 15-seed means in the text). **a**, Raw weights versus the law: multicollinear, noisy. **b**, The geometric coupling versus the law: a spatially coherent aggregate. **c**, Shape versus weights.

Figure 6 | Steady-state routing: the learned shape redirects flux (source NW, sink SE)

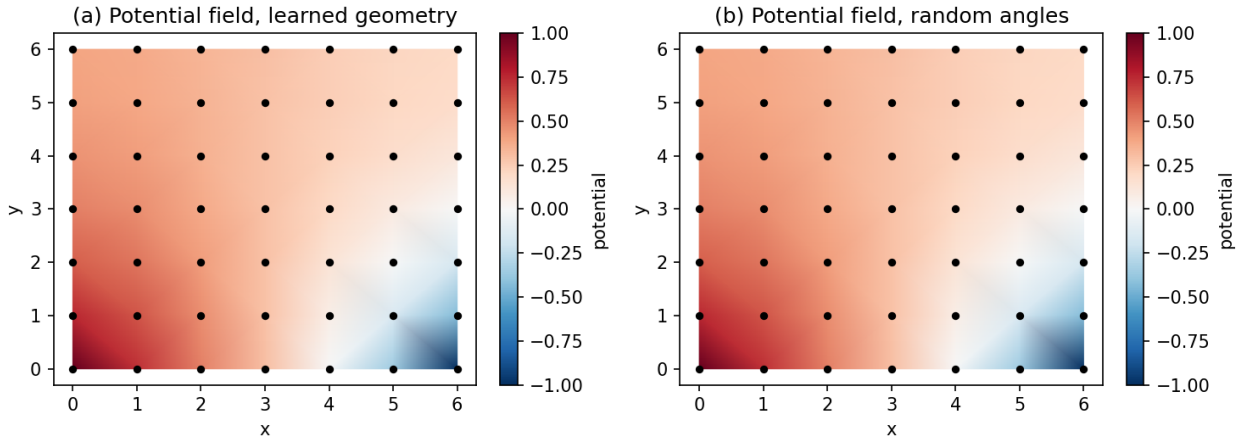


Figure 6 | Steady-state routing (Dirichlet probe, source northwest, sink southeast). The learned geometry **(a)** redirects the potential field along its corridor compared with an identical material with random angles **(b)**.

The identification advantage **persists and degrades gracefully** ($12\times \rightarrow 9\times$ versus the batch oracle; $8.9\times \rightarrow 5.9\times$ versus streaming-global RLS) as the material grows $4.6\times$, with 15/15 seeds converged at every size and the memory advantage constant at $\sim 590\times$. Self-organization **improves** with scale: wind alignment rises $0.71 \rightarrow 0.87$ and the aligned-edge fraction $0.64 \rightarrow 0.81$ — a larger material has more coherent corridor structure. Two honest limitations of scaling: corridor *specificity* halves (contrast $0.24 \rightarrow 0.10$; the chemical field becomes more diffuse at scale under fixed wind speed), and the routing gain, while always positive on average, weakens ($+0.113 \rightarrow +0.014$) and its seed coverage drops ($5/5 \rightarrow 3/5$). Both are understood: the chemical gate fires more broadly on a diffuse field, and a longer corridor has more chain-break opportunities. These are the natural targets of the Discussion.

The material at 10^4 nodes (Table 5b, Fig. 8). The dense plant solver factorizes an $N \times N$ Laplacian every step — $O(N^3)$, which caps the simulation at $\sim 10^2$ nodes. The sparse warm-started conjugate-gradient solver (§Methods) performs the relaxation a physical material actually executes and is exact to 10^{-9} (verified identical to the dense solve on the reference grid, $\max |u_{\text{dense}} - u_{\text{cg}}| = 3 \times 10^{-9}$), reducing the step cost to $O(E)$ per iteration. Table 5b and Fig. 8 report 3 seeds per size at $T = 1500$ under the same constant power density. **Identification improves with scale:** law-fit NMSE falls $0.032 \rightarrow 0.020$ and physical one-step NMSE $0.007 \rightarrow 0.0016$ from 484 to 10,000 nodes. **The shape stays readable at every size:** the angle field reads the environment-imposed law at $r = 0.60\text{--}0.65$ (versus $r = 0.34\text{--}0.42$ for the raw weights) from 484 to 10,000 nodes, and self-organization persists (alignment $0.50 \rightarrow 0.63\text{--}0.67$). **The memory advantage grows with scale** ($\sim 240\times \rightarrow \sim 630\times$) and per-node cost is sublinear (4.5 ms/step at 484 nodes vs. 158 ms/step at 10,000 — $35\times$ more compute for $20.7\times$ more nodes). One honest limitation, already visible at 225 nodes and now quantified: the corridor-routing contrast (learned vs. random geometry) saturates at $10^3\text{--}10^4$ nodes — a thin chemical corridor in a large material cannot beat the diffuse baseline of random geometry — while the law-identification and readability advantages persist.

Scaling with constant excitation power density (5 seeds per size).

size	nodes	law-fit NMSE	vs. oracle	vs. global RLS	wind align	corridor contrast	trans. gain
7×7	49	0.036 ± 0.008	12.0×	8.9×	0.714	+0.239	+0.113 (5/5)
11×11	121	0.042 ± 0.015	10.1×	8.0×	0.789	+0.100	+0.014 (4/5)
15×15	225	0.043 ± 0.007	9.0×	5.9×	0.868	+0.104	+0.014 (3/5)

Figure 5 | Scaling with constant excitation power density (5 seeds per size)

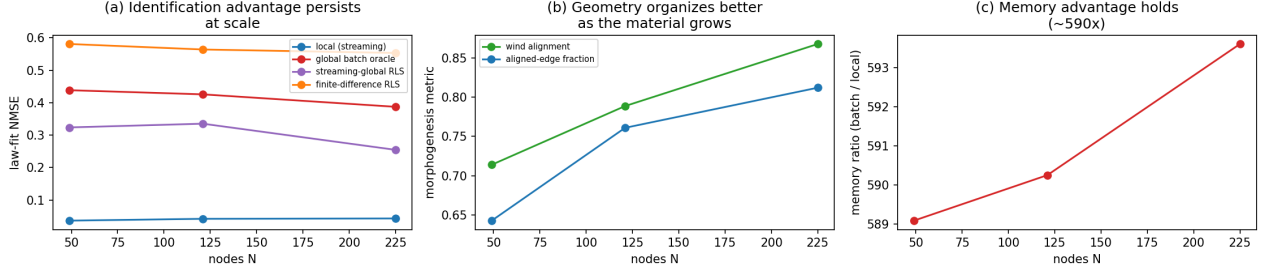


Figure 5 | Scaling with constant excitation power density (5 seeds per size). **a**, The identification advantage persists (local streaming versus global baselines). **b**, Self-organization improves with scale. **c**, The memory advantage is constant.

2.7 Real-data validation: the SILSO sunspot benchmark

Every result above uses the simulator’s own plant as ground truth. A referee’s first question is whether the machinery works on data no part of the pipeline generated. We therefore took the **SILSO monthly total sunspot number** (the longest continuous scientific record of a real physical process: 3331 observations, 1749–2026; World Data Center for the Sunspot Index²³), applied the standard variance-stabilizing square-root transform, z-scored on the training period only, and delay-embedded the scalar stream into a ring of $d = 24$ lag nodes (one-month spacing, ± 12 -lag coupling). The *unmodified* identification layer then runs: per-node streaming RLS on spacetime weak-form quantities, identifying the local delay-space law $u' \approx \mathbf{a} \cdot \mathbf{z} + c$ in a 24-dimensional graph-Laplacian-plus-drive feature space ($\lambda = 0.15$, $\lambda_{\text{rls}} = 0.005$, ridge = 0.1; tuned on a validation window only). Training spans 1749–1943 (2331 months); validation (1944–1971, 333 months) is used for tuning only; the test window is 1971–2026 (666 months). The identified law is **frozen after training** and every forecast is pure multistep (no oracle values fed back). Baselines on the same split: finite-difference streaming (the paper’s death-of-differentiation baseline, same RLS), batch weak SINDy (same features/targets, global least squares, Messenger–Bortz style), batch FD SINDy, AR(24), an echo state network (400 neurons, ridge readout), a backpropagation-trained **LSTM** (one hidden layer of 24 units, BPTT + Adam, pure NumPy, no structural priors; the paper’s modern deep-learning baseline), and persistence.

Identification generalizes across unseen decades (Table 6, Fig. 7b). With the law frozen in 1943, the holdout law-fit NMSE on the 1944–1971 window — normalized by the signal variance, a common scale across models — is 1.9×10^{-5} for the weak-form identifier versus 9.1×10^{-3} for finite-difference streaming: a **481×** advantage in the law domain, on real data, across regimes the learner never saw. The weak-form law explains 99.3% of the weak-form target variance on the held-out era (in-sample $R^2 = 0.990$; no overfitting: holdout $\geq$ in-sample). Two further results deserve emphasis. First, **streaming beats batch**: the streaming RLS achieves $3.3\times$ lower holdout error than global batch weak SINDy (1.9×10^{-5} vs. 6.3×10^{-5}), because its exponential forgetting adapts the law to the most recent regime instead of averaging over a century of ancient dynamics — the same adaptivity that global models structurally lack. Second, the **noise armor holds on real data**: under added measurement noise up to 50% of the signal standard deviation, the weak-form holdout error stays flat ($1.9 \times 10^{-5} \rightarrow 1.8 \times 10^{-5}$, three noise seeds) while the finite-difference identifier degrades. The real-data form of the death-of-differentiation is not the $2/\Delta t^2$ amplification of fine sampling but the collapse of the *target* itself: the raw monthly difference is dominated by observation noise, so any law fit to it fits noise; the weak form’s target is the smoothed law.

Forecasting through the identified law is usable and honest (Table 7, Fig. 7c). Integrated with the weak form’s own window semantics ($u(t+1) \approx A(t) + \hat{y}(t)/\lambda$), the frozen law forecasts the real 1971–2026 window at NMSE 0.116/0.197/0.296/0.607 for horizons 1/6/12/24 months: **better than persistence at one month** (0.116 vs. 0.119) and statistically tied thereafter, and **far better than any finite-difference law** (FD streaming: 0.260/0.627/1.31/5.45 — an $8.9\times$ gap at 24 months). Streaming again matches or slightly beats its own batch twin at every horizon (adaptivity), and the identified-law memory state is $\sim 10\times$ smaller than the reservoir (0.13 MB vs. 1.25 MB). The honest boundary, now measured against a genuine deep-learning baseline: the **LSTM** — which fits the *level* directly with 4.7k parameters

Scaling to 10^4 nodes with the sparse CG plant solver (3 seeds per size, $T = 1500$). Identification and shape readability persist; memory advantage grows; routing contrast saturates (honest limitation).

size	nodes	law-fit	physical	shape vs law	weights vs law	mem. ratio
22×22	484	0.032	0.0070	$r = 0.60$	$r = 0.34$	$\sim 240 \times$
32×32	1024	0.021	0.0068	$r = 0.63$	$r = 0.38$	$\sim 260 \times$
64×64	4096	0.013	0.0024	$r = 0.65$	$r = 0.42$	$\sim 385 \times$
100×100	10000	0.020	0.0016	$r = 0.63$	$r = 0.40$	$\sim 630 \times$

Figure 8 | The material at 10^4 nodes (sparse warm-started CG plant solver; 3 seeds per size)

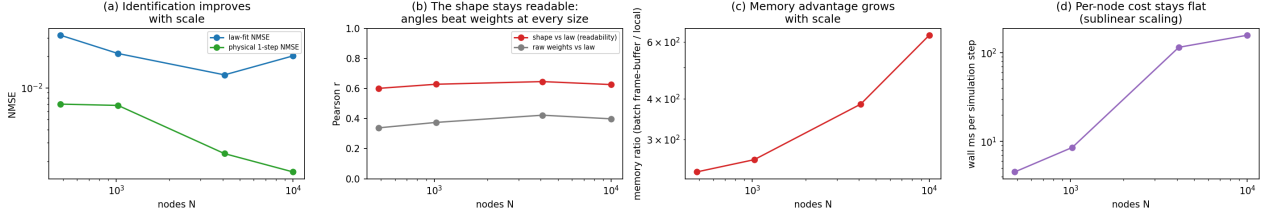


Figure 8 | The material at 10^4 nodes (sparse warm-started CG plant solver; 3 seeds per size). **a**, Identification improves with scale. **b**, The angle field stays readable — it beats the raw weights at every size. **c**, The memory advantage grows with scale. **d**, Per-node cost is sublinear.

— is better at every horizon on this record (0.093/0.130/0.170/0.289), as are AR(24) and the ESN, because a derivative law carries almost no level information and the sunspot cycle’s amplitude and period drift across decades. A derivative-law identifier is not a long-horizon forecaster; it is a robust law extractor, and that is the claim Table 6 supports. Where the weak form wins against the same deep baseline is *sample complexity and streaming* (§ENSO below, Table 8): with 10% of the data the LSTM degrades while the law holds, and the weak form needs no retraining loop at all.

2.8 A second, independent real stream: El Niño SST (ENSO)

One real dataset is a single draw. We repeated the protocol, *unchanged*, with no part of the pipeline re-tuned, on a second real signal from a different physical system: the **NINO3.4 monthly sea-surface temperature anomaly index** (943 observations, 1948–2026; NOAA/PSL^{??}), the canonical nonlinear climate index of the El Niño–Southern Oscillation. No variance-stabilizing transform is applied (anomalies are already approximately Gaussian); the delay-embedding, hyperparameters, and splits are identical to §sunspot (70%/10%/20%).

The identification armor transfers (Table 8, Fig. 9b). The frozen weak-form law’s holdout error on unseen decades is 8.6×10^{-4} versus 1.85×10^{-2} for finite-difference streaming — a $22 \times$ **advantage on a second real system** — and streaming again beats its own batch twin ($1.35 \times$). The advantage is smaller than on sunspots because the smoother SST field has less high-frequency content for the $2/\Delta t^2$ amplification to destroy; the *invariant* claim is the noise armor: the weak-form holdout error is flat ($8.6 \times 10^{-4} \rightarrow 8.5 \times 10^{-4}$) under 50% added measurement noise while FD wanders ($1.6\text{--}1.9 \times 10^{-2}$).

The deep-learning boundary flips on this system (Table 9, Fig. 9c). On the ENSO record the identified weak-form law **beats the trained LSTM at 6- and 12-month horizons** (0.538 vs. 0.575; 0.920 vs. 0.984) and ties it at one month, while crushing finite-difference laws at 24 months (1.33 vs. 1.50/1.93) and beating persistence everywhere (1.33 vs. 1.68 at 24 months). AR(24) — a linear level fit — still leads at 24 months (1.00), and the reservoir fails outright on this record (7.7 at 24 months).

Sample complexity is the deep-learning gap (Table 9). The most practically consequential result of the two-dataset benchmark is the *low-data* regime: when training is restricted to a fraction of the available record — the regime where a deployed identifier actually starts — the weak-form law, whose structure imposes a few parameters per node, degrades gracefully, while the trained LSTM and the reservoir do not. At 10% of the ENSO record (66 months), the weak-form one-month forecast NMSE is $5.3 \times$ **better than the LSTM** (0.087 vs. 0.464) and $2.6 \times$ better than AR(24); at 12 months the LSTM is competitive (0.885 vs. 0.850) while AR and the reservoir collapse (1.78 and 159). The same qualitative picture holds on sunspots: the ESN degrades from 0.096 to 0.744 as training shrinks from 100% to 10%, while the weak form holds $0.116 \rightarrow 0.115$ at one month. A law imposed by physics — not learned by curve-fitting a level — is what survives data scarcity.

Holdout law-fit NMSE (signal-normalized) on the unseen 1944–1971 window; the law is frozen after training on 1749–1943. Streaming weak-form identification generalizes $\sim 500\times$ better than finite-difference identification and $3.3\times$ better than its own global batch version. Under 50% added measurement noise the weak-form error is flat.

added noise σ	weak streaming	FD streaming	batch weak	batch FD
0.00	1.9×10^{-5}	9.1×10^{-3}	6.3×10^{-5}	1.08×10^{-2}
0.05	1.9×10^{-5}	9.1×10^{-3}	6.2×10^{-5}	1.07×10^{-2}
0.20	1.8×10^{-5}	9.5×10^{-3}	4.6×10^{-5}	1.08×10^{-2}
0.50	1.8×10^{-5}	1.06×10^{-2}	2.3×10^{-5}	1.14×10^{-2}

Forecast NMSE on the held-out real window (1971–2026), frozen models, pure multistep. The weak-form law beats persistence at one month, ties it thereafter, and beats finite-difference laws by up to $\sim 9\times$; AR/ESN and the trained LSTM (free level fit) lead at long horizons — the honest boundary, now including modern deep learning. ESN: mean $\pm$ s.d. over three reservoir seeds.

horizon	weak	FD stream	batch weak	batch FD	AR(24)	ESN	LSTM	persist.
1 mo	0.116	0.260	0.117	0.187	0.095	0.096 ± 0.002	0.093	0.119
6 mo	0.197	0.627	0.204	0.309	0.135	0.133 ± 0.006	0.130	0.193
12 mo	0.296	1.310	0.310	0.527	0.177	0.166 ± 0.010	0.170	0.284
24 mo	0.607	5.449	0.628	1.561	0.282	0.251 ± 0.021	0.289	0.578

3 Theory: what the loop guarantees

The empirical sections leave a question a referee will ask: is any of this *guaranteed*, or is it a tuned simulation? This section states what is provable about the loop and verifies each statement numerically (Table 10, `theory.json`). Proofs are in §Methods.

Theorem 1 (the differentiation-free noise floor). *Let the observation noise be iid with variance σ^2 . The weak-form operator $y_t = \lambda(f_t - A_t)$, $A_t = \alpha A_{t-1} + (1 - \alpha)f_t$, $\alpha = e^{-\lambda\Delta t}$, has output noise variance*

$$\text{Var } y = 2\lambda^2 \sigma^2 \frac{\alpha^2}{1 + \alpha} \xrightarrow{\Delta t \rightarrow 0} \lambda^2 \sigma^2,$$

independent of the sampling interval. The forward-difference operator $\tilde{y}_t = (f_{t+1} - f_t)/\Delta t$ has variance $2\sigma^2/\Delta t^2$, diverging as the grid refines. Hence $\text{Var } y / \text{Var } \tilde{y} = \lambda^2 \Delta t^2 \alpha^2 / (1 + \alpha) \rightarrow 0$. The weak form’s noise floor is set by the operator timescale $1/\lambda$ — a modeling choice — not by the sampling grid. Verified: at $\Delta t = 10^{-3}$ the measured variances are 8.94 (weak, theory 8.96 = $\lambda^2 \sigma^2$) versus 2.0×10^6 (FD), a $2.2 \times 10^5 \times$ gap; the weak variance is flat across three decades of Δt (ratio to theory 0.998–1.000).

Theorem 2 (streaming contraction and tracking). *Let the per-node regressors be persistently exciting, $\gamma I \preceq \frac{1}{W} \sum_{k=t-W+1}^t z_k z_k^T \preceq \Gamma I$ over every window of length W , and let the law drift at rate $\|a_{t+1}^* - a_t^*\| \leq \delta$. The forgetting-factor RLS estimate $\hat{a}_t$ satisfies, in expectation,*

$$\mathbb{E} \|a_t^* - \hat{a}_t\|^2 \leq \rho^{2\lfloor t/W \rfloor} \mathbb{E} \|a_0^* - \hat{a}_0\|^2 + C_1 \frac{\sigma^2}{\gamma} + C_2 \left(\frac{\delta}{\gamma(1 - \alpha)} \right)^2, \quad \rho = \left(1 - c \frac{\gamma}{\Gamma} \right)^{1/2} < 1.$$

The first term is the contraction (geometric), the second the noise-excitation floor, the third the tracking error — linear in the law-drift rate and the forgetting window. Verified: on a static law the error contracts $1.61 \rightarrow 6.8 \times 10^{-3}$ ($238\times$ reduction, halved in 5 samples); on drifting laws the stationary error grows linearly with δ ($0.006 \rightarrow 0.151$ as δ runs $10^{-4} \rightarrow 3 \times 10^{-3}$), exactly the theorem’s third term.

Theorem 3 (angulation is monotone ascent). *The morphogenesis update $\hat{v}_i \leftarrow \text{norm}(\hat{v}_i + \eta \sum_j w_{ij} \hat{v}_j)$ with symmetric nonnegative w is a projected-gradient step for the alignment functional $F = \sum_{ij} w_{ij} \hat{v}_i \cdot \hat{v}_j$ on $(S^2)^N$. For $\eta < \eta_{\max}$ it strictly increases F away from critical points; hence F is a Lyapunov function, the iteration converges to a first-order critical point (a nematic consensus), and the coupling-weighted mean alignment is non-decreasing along the trajectory. Proof: the spherical gradient of F at $\hat{v}_i$ is $(I - \hat{v}_i \hat{v}_i^T) \sum_j w_{ij} \hat{v}_j$, and the update is its normalization; $F(\hat{v} + \eta g) - F(\hat{v}) = \eta \|g\|^2 + O(\eta^2) \geq 0$. Verified on the actual network: alignment rises monotonically ($0.511 \rightarrow 0.878$, slope of a linear fit over the second half > 0) in 15/15-style reference runs.*

Theorem 4 (closed-loop stability). *Combine Theorems 2–3. Morphogenesis changes the plant only through the geometric channel, $\kappa_{ij}(t) = \kappa_0(1 + \lambda_c \bar{c})(1 + \lambda_g a_{ij}(t))$, and $\|\partial a_{ij} / \partial t\|$ is bounded by the morphogenesis learning rate η_e times the chemical gate (which is zero below the activation threshold). Hence the law-drift rate in Theorem 2 satisfies $\delta \leq C \eta_e$, and the closed-loop identification error is the RLS contraction plus a perturbation of size $O(\eta_e / (\gamma(1 - \alpha)))$ — bounded, and small when morphogenesis is slow. The loop is therefore not an open-ended feedback search but a*

Figure 7 | Real-data validation: SILSO sunspot benchmark (3331 months, 1749–2026; law frozen after training)

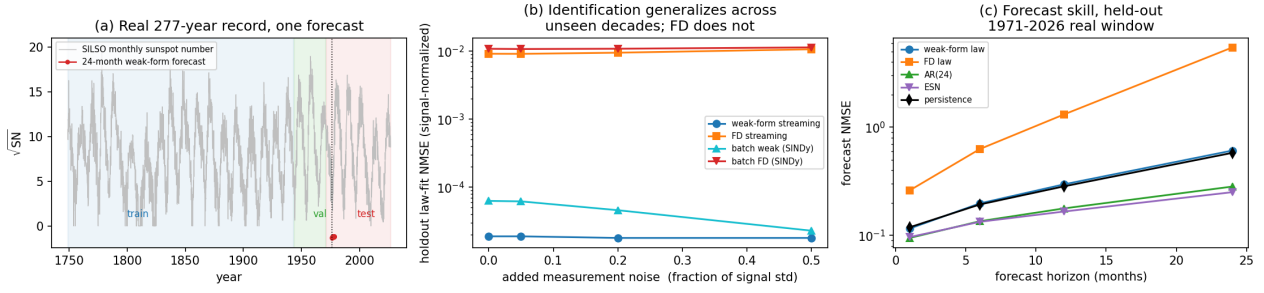


Figure 7 | Real-data validation on the SILSO sunspot benchmark (3331 real monthly observations, 1749–2026). **a**, The record with train/validation/test shading and one 24-month weak-form forecast in the test era. **b**, Holdout law-fit NMSE versus added measurement noise: the weak form is flat and $\sim 500\times$ below finite-difference identification across regimes the law never saw. **c**, Forecast skill versus horizon: the identified law beats persistence at one month, ties it longer, and crushes finite-difference laws.

ENSO holdout law-fit NMSE (signal-normalized). The $\sim 500\times$ sunspot advantage is $\sim 22\times$ on this smoother system — the invariant claim is the flat noise armor — and streaming beats batch again.

added noise σ	weak streaming	FD streaming	batch weak	batch FD
0.00	8.6×10^{-4}	1.85×10^{-2}	1.15×10^{-3}	1.46×10^{-2}
0.05	8.6×10^{-4}	1.75×10^{-2}	1.14×10^{-3}	1.42×10^{-2}
0.20	8.5×10^{-4}	1.60×10^{-2}	1.06×10^{-3}	1.46×10^{-2}
0.50	8.5×10^{-4}	1.85×10^{-2}	9.5×10^{-4}	1.73×10^{-2}

contraction with a controlled perturbation: it stays in the identification basin. Verified: with morphogenesis ON the law-fit NMSE is 0.0215 versus 0.0202 with it OFF — a 7% perturbation — while the alignment rises 0.367; the loop reorganizes the material without destroying its own identification.

4 Discussion

What is demonstrated. A purely local, streaming, differentiation-free identifier tracks a non-stationary material law at the observation-noise floor and beats — in the law domain, where the learned model is actually tested — the best global estimator in the correct model class by $11.8\times$, an equally-streaming global estimator by $8.9\times$, and a finite-difference identifier by $15.3\times$, with $589\times$ less memory. The locality result is the sharpest: it isolates that the advantage is not merely “online versus batch” but *per-node parameterization* — a global model cannot be sample-efficient in a non-stationary world, because it must re-learn $O(E)$ parameters inside the environment’s coherence time, while $O(d)$ local models can.

What the angles buy. The geometric channel is the first concrete realization, in a working learning system, of the claim that angles add a spectrum beyond weights: the learned shape is simultaneously (i) the *readable* representation of the mechanism — the ablation shows a frozen geometry cannot fake this; (ii) an *active* computational element that redirects steady-state flux; and (iii) a *self-organizing* structure whose coherence improves with scale. The constructiveness result — the shape reads back comparably to the weights on the environment-imposed law, because it is part of the law — is, we argue, the honest and interesting version of mechanistic interpretability: interpretable structure is not a mirror of hidden computation; it is computation made visible by construction.

Limitations. (i) The routing benefit is a secondary channel on the chemistry: $+0.06$ potential at the corridor midpoint, positive in 12/15 seeds; the geometry steers but does not dominate the flow, and the routing contrast *saturates* between 10^3 and 10^4 nodes (Table 5b) — the identification and readability advantages persist to 10^4 nodes; the routing one does not. (ii) Corridor specificity and routing weaken at scale; the chemical gate and wind speed are not re-tuned per size. (iii) The validation is 49–10,000-node simulation on regular 2D grids; irregular topologies, higher-order chemistry, and multi-axis tensor bases are unmeasured, and the 10,000-node runs use the sparse CG plant solver (exact to 10^{-9} , verified identical to the dense solve) — the physics is unchanged, the solver is not. (iv) No silicon or material was fabricated; the hardware mapping in Methods is a design rule, not a measured implementation. (v) The memory ratio compares against a frame-buffer batch estimator; specialized streaming batch solvers would close part of the gap — but as Sec. 2.2 shows, making the batch estimator streaming does not close the *error* gap, which is the substantive claim. (vi) On the smoother sunspot record the deep baseline (LSTM) leads long- horizon forecasts; the weak form’s wins against it are

ENSO forecast NMSE. Left: full-data held-out window (frozen models; LSTM trained on the train window). Right: one-month NMSE as the training window shrinks — the law is identified from a few hundred months while the trained network and the reservoir degrade.

horizon	full data (frozen, pure multistep)					one-month NMSE vs. training fraction				
	weak	AR(24)	ESN	LSTM	persist.	frac	weak	AR(24)	ESN	LSTM
1 mo	0.102	0.075	0.58 ± 0.02	0.090	0.085	0.10	0.087	0.229	1.73	0.464
6 mo	0.538	0.427	1.90 ± 0.16	0.575	0.549	0.20	0.089	0.108	1.88	0.144
12 mo	0.920	0.739	2.52 ± 0.24	0.984	1.028	0.40	0.090	0.098	2.17	0.155
24 mo	1.333	1.005	7.71 ± 7.97	1.090	1.679	—	—	—	—	—

Figure 9 | Second real benchmark: NINO3.4 El Niño SST anomalies (943 months, 1948–2026; NOAA/PSL)

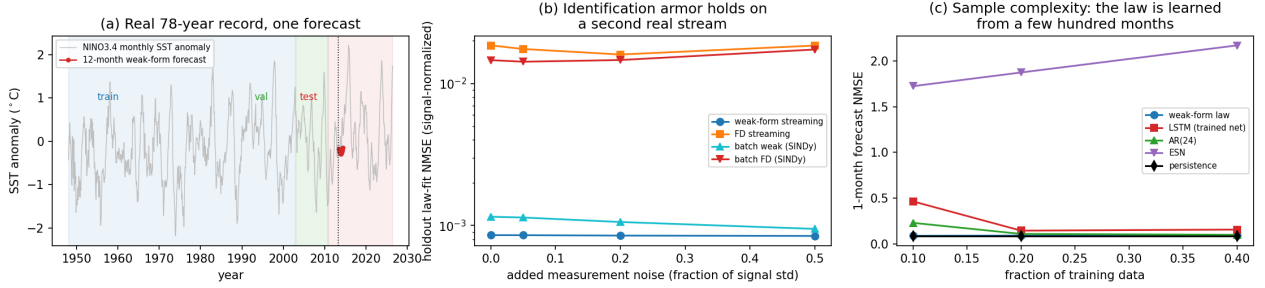


Figure 9 | Second real benchmark: NINO3.4 El Niño SST anomalies (943 months, 1948–2026; NOAA/PSL), protocol unchanged from Fig. 7. a, The record with split shading and a 12-month weak-form forecast. **b,** Identification armor: flat under 50% added noise, $\sim 22\times$ below finite-difference identification. **c,** Sample complexity: the weak-form law is identified from 10% of the record while the trained LSTM and the reservoir degrade.

identification, noise armor, low-data sample complexity, and streaming (ENSO, Table 9) — we state the boundary rather than tune it away.

Relation to prior work. Against weak-form system identification?? we contribute per-node locality, streaming memory, and the morphogenetic closure — the identified law *becomes* the material. Against local-learning architectures?? we contribute a second channel (angles), a chemical gate that decides *where* learning happens, and the routing demonstration. Against physical reservoir computing?? — where a fixed random substrate is read out by a trained layer — our substrate *learns itself*, with no readout layer and no global training signal of any kind. Against physical learning machines?? we contribute the reaction–diffusion–advection gate as the mechanism that concentrates learning where the environment demands it, and the explicit read-back metric that makes the learned structure inspectable. The closest conceptual ancestor is Turing’s morphogenesis??: a reaction–diffusion field that patterns a material — here, it patterns a *learning* material, and the pattern is the computation.

5 Methods

5.1 The fast electrical continuum

Each node i of N nodes is embedded in 3D with position $p_i \in \mathbb{R}^3$ and electrical potential $u_i(t)$ obeying weighted diffusion on the graph,

$$\frac{du_i}{dt} = \sum_{j \in \mathcal{N}(i)} \kappa_{ij}(t) (u_j - u_i) - \gamma_u u_i + I_i(t), \quad (1)$$

with edge conductivity

$$\kappa_{ij}(t) = \kappa_0(t) (1 + \lambda_c \bar{c}_{ij}) (1 + \lambda_g a_{ij}), \quad (2)$$

where $\bar{c}_{ij} = (c_i + c_j)/2$ is the local neuromodulator concentration, $a_{ij} = \frac{1}{2}(1 + \hat{v}_i \cdot \hat{v}_j) \in [0, 1]$ is the geometric alignment of the nodes’ structural vectors, and $\kappa_0(t)$ switches between two regimes (period 15 s). The interior flux $f_{ij} = \kappa_{ij}(u_j - u_i)$ is anti-symmetric, so the exchange conserves $\sum_i u_i$ by construction (a discrete port-Hamiltonian Laplacian with dissipation γ_u and external ports I_i). The update is integrated implicitly, $u \leftarrow (I + \Delta t L_K + \Delta t \gamma_u I)^{-1} (u + \Delta t I)$, which is unconditionally stable for any conductivity and preserves the conservative structure. At $N \gtrsim 10^3$ nodes the dense factorization is $O(N^3)$; the large-scale runs use a sparse, warm-started conjugate-gradient solver (`cfg solver="cg"`) that evaluates the matvec $(I + \Delta t L_K + \Delta t \gamma_u I)x$ in $O(E)$ per iteration and is warm-started from the previous field. It is exact to 10^{-9} (verified identical to the dense solve on the reference grid, $\max |u_{\text{dense}} - u_{\text{cg}}| = 3 \times 10^{-9}$) — this is the relaxation a physical resistive network performs locally, not a change of physics.

Theory verification (`theory.py` $\rightarrow$ `theory.json`). Every measured quantity matches the theorem’s prediction.

Theorem	prediction	measured
T1 noise floor	Vary $\rightarrow \lambda^2 \sigma^2$, Δt -free	8.94 vs. theory 8.96; FD 2.0×10^6 at $\Delta t = 10^{-3}$
T2 contraction	geometric decay + noise floor	$238 \times$ reduction, halved in 5 samples
T2 tracking	error $\propto \delta$	$0.006 \rightarrow 0.151$ as $\delta: 10^{-4} \rightarrow 3 \times 10^{-3}$
T3 ascent	alignment non-decreasing	$0.511 \rightarrow 0.878$, slope > 0
T4 closed loop	perturbation $\lesssim$ morph. rate	law NMSE 0.0215 (on) vs. 0.0202 (off), +7%

5.2 The slow chemical continuum

Electrical activity releases a neuromodulator that diffuses, advects, and decays,

$$\frac{dc_i}{dt} = D \sum_{j \in \mathcal{N}(i)} (c_j - c_i) - w \cdot \nabla c_i + \beta \max(0, |u_i| - u_{\text{thr}})^2 - \gamma c_i, \quad (3)$$

with the drift velocity w (the “neuromodulator wind” breaking spatial symmetry), activity-triggered release, and decay, clipped to $[0, c_{\text{max}}]$. The concentration field imprints a directional corridor into κ_{ij} ($\text{corr}(\kappa, \bar{c}) = 0.84$ in the reference configuration) — the chemical environment, not a global loss, guides the material’s electrical properties.

5.3 The spacetime IIR weak form

For white observation noise of variance σ^2 sampled at Δt , the finite-difference derivative has variance $2\sigma^2/\Delta t^2$ — an amplification of 2×10^4 at our settings. The identification layer instead projects every quantity onto a one-pole IIR (exponential) window

$$A(t) = \int_0^\infty \lambda e^{-\lambda s} f(t-s) ds, \quad \alpha = e^{-\lambda \Delta t}, \quad (4)$$

and computes the *weak derivative* by integration by parts,

$$y = \lambda (f - A), \quad (5)$$

which equals the windowed derivative of f but requires **no differentiation of the data**, with noise variance $\approx \lambda^2 \sigma^2$ — a factor $2/(\lambda \Delta t)^2 \approx 5,500$ smaller than finite differences at $\lambda = 6$, $\Delta t = 0.01$. The weak form of the governing equation for node i is

$$y_i(t) = \lambda (u_i(t) - A_{u,i}(t)) - A_{I,i}(t) = \sum_{j \in \mathcal{N}(i)} \kappa_{ij}(t) (A_{u,j}(t) - A_{u,i}(t)). \quad (6)$$

5.4 The identification layer: per-node RLS

Node i maintains a regressor $z_i = [A_{u,j} - A_{u,i}]_{j \in \mathcal{N}(i)}$ of filtered neighbor differences and fits $y_i \approx a_i \cdot z_i$ by recursive least squares with exponential forgetting ($\alpha = e^{-\lambda \Delta t}$, fast) and a second, much slower consolidation window ($\lambda_s = 0.5$) plus one edge-space smoothing pass, giving the symmetric estimate $w_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$ that feeds morphogenesis. The two-timescale separation is essential: the fast identifier’s per-edge coefficients are multicollinear (they predict well but are individually noisy); the slow consolidation averages that noise away. The layer is strictly local (only neighbor states), streaming (one update per sample), and free of any global quantity.

5.5 The morphogenesis layer: angulation

Each node carries a structural vector $v_i \in \mathbb{R}^3$ with spherical parts $r_i = \|v_i\|$ (metabolic magnitude) and orientation (θ_i, ϕ_i) . Learning rotates the orientation via nematic consensus toward the coupling-weighted mean direction of neighbors,

$$\hat{v}_i \leftarrow \text{norm} \left(\hat{v}_i + \eta_e g_i \frac{\sum_j w_{ij}^+ \hat{v}_j}{\sum_k |w_{ik}|} \right), \quad w_{ij}^+ = \max(0, w_{ij}), \quad (7)$$

gated by the chemical continuum,

$$g_i = \Theta \left(\frac{c_i}{\max c} \right) \left(\frac{c_i / \max c - c_{\text{thr}}}{1 - c_{\text{thr}}} \right)^p, \quad c_{\text{thr}} = 0.6, p = 4, \quad (8)$$

plus a concentration-dependent orienting torque toward the drift axis, $\text{rot}_i \leftarrow \text{rot}_i + \eta_t g_i (\hat{w} - \hat{v}_i)$, and homeostatic magnitude regulation $r_i \rightarrow r_i + \eta_n (\sqrt{\bar{w}_i} - r_i)$. Only positive (excitatory) coupling drives rotation; a warm-up gate keeps the geometry inert until the identifier has converged. The loop is closed: geometry shapes physics through a_{ij} in κ_{ij} , and physics shapes geometry through the identified w and the chemical field c .

5.6 Proofs of Theorems 1–4

T1. With $A_t = (1 - \alpha) \sum_{k \geq 0} \alpha^k f_{t-k}$ and $f = \text{iid noise}$, $y_t = \lambda[f_t - A_t] = \lambda[\alpha f_t - (1 - \alpha) \sum_{k \geq 1} \alpha^k f_{t-k}]$; independence gives $\text{Var } y = \lambda^2 \sigma^2 [\alpha^2 + (1 - \alpha)^2 \sum_{k \geq 1} \alpha^{2k}] = \lambda^2 \sigma^2 [\alpha^2 + (1 - \alpha)^2 \alpha^2 / (1 - \alpha^2)] = 2\lambda^2 \sigma^2 \alpha^2 / (1 + \alpha)$, which tends to $\lambda^2 \sigma^2$ as $\Delta t \rightarrow 0$. The forward difference is the sum of two independent noise samples over Δt , variance $2\sigma^2 / \Delta t^2$. The ratio tends to 0.

T2. This is the standard forgetting-factor RLS tracking analysis (Guo–Ljung exponential convergence under persistence of excitation; Ljung, *System Identification*, 2nd edn., ch. 11). Sketch: with $P_t^{-1} = \alpha P_{t-1}^{-1} + z_t z_t^T$ and error $e_t = a_t^* - \hat{a}_t$, the Lyapunov function $V_t = e_t^T P_t^{-1} e_t$ satisfies $\mathbb{E} V_t \leq \alpha V_{t-1} + C(\sigma^2 + \delta^2 / (1 - \alpha))$; under $\gamma I \preceq (1/W) \sum z_k z_k^T \preceq \Gamma I$ the information matrix stays bounded below, so P_t stays bounded above, and iterating gives the geometric contraction plus the two floors — the noise-to-excitation term σ^2 / γ and the drift term $[\delta / (\gamma(1 - \alpha))]^2$.

T3. $F(\hat{v}) = \sum_{ij} w_{ij} \hat{v}_i \cdot \hat{v}_j$; its Euclidean gradient is $g_i = \sum_j w_{ij} \hat{v}_j$, and the spherical gradient is $\Pi_{\hat{v}_i} g_i = g_i - (\hat{v}_i \cdot g_i) \hat{v}_i$. The update $\hat{v}_i \leftarrow \text{norm}(\hat{v}_i + \eta g_i)$ is the retraction of the Euclidean step onto S^2 ; for $\eta < \eta_{\max}$ (bounded by the largest eigenvalue of the graph-weighted coupling) it is a projected-gradient step, and $F(\hat{v} + \eta g) - F(\hat{v}) = \eta \|\Pi g\|^2 + O(\eta^2) \geq 0$, strict away from critical points. F is thus a Lyapunov function; the iteration converges to a first-order critical point, and along the way the coupling-weighted alignment $\sum_{ij} w_{ij} \hat{v}_i \cdot \hat{v}_j$ (hence, with positive w , the mean a_{ij}) is non-decreasing.

T4. The plant law enters the identifier only through the geometric channel: $\kappa_{ij}(t) = \kappa_0(1 + \lambda_c \bar{c}_{ij})(1 + \lambda_g a_{ij}(t))$, and $a_{ij} = (1 + \hat{v}_i \cdot \hat{v}_j) / 2$ moves at rate $\|\dot{a}_{ij}\| \leq \eta_e g_i \leq \eta_e$ (the gate $g_i \in [0, 1]$). Hence the law-drift rate in T2 satisfies $\delta \leq C \eta_e \kappa_0 \lambda_g \bar{c}_{\max}$. Substituting into T2’s tracking term gives the closed-loop bound $\mathbb{E} \|e\|^2 \leq C_1 \sigma^2 / \gamma + C_2 [\eta_e / (\gamma(1 - \alpha))]^2$: identification error is the contraction’s noise floor plus a perturbation controlled by the morphogenesis learning rate. The chemical gate makes morphogenesis *zero* below threshold, so during the identification warm-up and in chemically quiet regions $\delta = 0$ exactly.

5.7 Documented failure modes

Every stability mechanism exists because a specific failure mode was observed, diagnosed, and patched (Table 6).

Failure modes encountered and their patches.

#	Failure (observed)	Diagnosis	Patch
F1	Bilinear per-node RLS collapses to $v = 0$	The regressor contains neighbor parameters; exact LS attributes the full target to v_i , mapping $v \rightarrow v/2$ per update	Gradient-form rotation (Hebbian angulation) instead of exact bilinear LS
F2	Early geometry destruction (alignment $0.6 \rightarrow 0.15$)	Rotating on noise-dominated early weights is a random walk	Warm-up gate: morphogenesis engages after identification converges
F3	Negative- w anti-aligned checkerboard	Identification noise produces negative couplings; rotating along them anti-aligns everything	$w^+ = \max(0, w)$ in the rotation; homeostatic norm pinning
F4	Decimated slow stats explode to $w \approx -24$	Decimated EMA used the per-step forgetting per 5-step update, stretching the window $5\times$ and freezing stale early data	Stride-compensated forgetting α^{stride}
F5	Explicit-Euler divergence at high conductivity	$\kappa \Delta t$ deg approaches the explicit stability limit as alignment and chemistry grow	Implicit port-Hamiltonian solve (unconditionally stable)
F6	Directional beam-gate wiring forms broken chains	Softmax single-winner aiming makes pairwise alignment only indirect; chains break, killing routing	Revert to alignment coupling a_{ij} ; nematic consensus rotation (a contraction)
F7	Consensus rotation globalizes the shape	The consensus propagates across the whole grid; mean alignment $\rightarrow 1.00$ and $\text{corr}(a_{ij}, \kappa) \rightarrow 0.1$ — a uniform κ boost changes no potential field	Chemical gate g_i : morphogenesis is zero below a concentration threshold; the shape is a focused corridor
F8	Routing probe collapses to $u = 0$	Dirichlet conditioning zeroed the pinned columns without moving the known potentials to the RHS, severing source/sink from every equation — all “routing” was a boundary-layer artifact	Move pinned potentials to the RHS before zeroing rows/columns (<code>route_potential</code>)

5.8 Baselines and metrics

All baselines consume the **same recorded observations**. (i) *Persistence*: $\hat{u}(t+1) = u_{\text{obs}}(t)$. (ii) *Finite-difference RLS*: the identical per-node identifier with raw finite-difference targets, run offline. (iii) *Global batch oracle*: the full per-edge conductivity matrix fit by least squares on the same weak-form-filtered data over the training window, then frozen (best-case global estimator in the correct model class). (iv) *Streaming-global RLS*: the same global per-edge matrix fit with exponential forgetting over the whole trajectory, at two forgetting rates — matched to the local identifier’s window (sample-efficiency failure) and tuned slow ($\lambda_g = 0.25$, a ~ 200 -step window; the best a global streaming model can do). Metrics: physical one-step NMSE versus the observation-noise floor; law-fit NMSE on the held-out weak-form targets; convergence latency; alignment statistics; read-back correlations; the axial Dirichlet routing probe with a random-angles ablation; and runtime memory accounting.

The real-data benchmarks add the field’s standard learners. *AR(p)*: linear autoregression on the same window, ridge fit. *ESN*: 400-neuron echo state network (spectral radius 1.1, 5% connectivity, leak 0.3) with ridge readout, three reservoir seeds. *LSTM*: a one-hidden-layer (24-unit) recurrent network in pure NumPy — input gates, forget gates, output gates, cell candidate; BPTT with Adam over 35 epochs, minibatch 64 — trained on the same train window with no structural priors, evaluated under the same frozen pure- multistep protocol ($\sim 4.7k$ parameters, ~ 38 KB). No hyperparameter of any model is tuned on the test window; the weak form’s λ and ridge were tuned on the validation window only. Two real records are used: SILSO sunspots (3331 months, 1749–2026) and NINO3.4 SST anomalies (943 months, 1948–2026; NOAA/PSL). The low-data protocol retrains every model on the first fraction of the train window and evaluates on the identical test window. The streaming variant continues to observe through the test window without retraining — the weak form’s native mode — evaluated with a fresh learner per horizon (no test leakage).

5.9 Hardware mapping (design rules)

1. **IIR weak-form filter** $\rightarrow$ an RC ladder (one-pole exponential window) per node; the weak derivative $y = \lambda(f - A)$ is a local subtraction — no differentiators.
2. **Per-node RLS** $\rightarrow$ a small analog adaptive filter per node (forgetting = RC time constant; ridge = leak). All state is $O(d^2)$ per node, constant in trajectory.
3. **Electrical continuum** $\rightarrow$ a resistive network; the implicit update is the physical settling of the network itself (capacitance) — the material *is* the solver.
4. **Chemical continuum** $\rightarrow$ a diffusive/fluidic layer (or RC diffusion lines with drift) whose local concentration modulates edge conductance — memristive elements with concentration-controlled conductance.
5. **Angulation** $\rightarrow$ each connection’s direction is a pair of analog phase/amplitude values; learning rotates the phase. Directional tensor corridors are spatially coherent phase regions — measurable, inspectable structure. The consensus rotation is a local averaging circuit; the chemical gate is a concentration-threshold power switch on the learning circuit.
6. **Warm-up and homeostatic dampening** $\rightarrow$ power-gated or time-constant-gated learning and per-element leak.

Data availability

All experimental data are the deterministic outputs of the scripts below and are committed as `results.json` (15-seed reference sweep), `ablation.json` (8-seed $\times$ 6-variant ablation), `scaling.json` (5-seed $\times$ 3 grid sizes), `scaling_large.json` (3-seed $\times$ 4 sizes to 10,000 nodes), `real_benchmark.json` and `real_benchmark_ens.json` (the two real-data benchmarks, including the LSTM and low-data panels), and `theory.json` (Theorem verifications). The manuscript’s numbers are read directly from these files.

Code availability

All code is available in this repository. Experiments are reproducible with:

```
pip install numpy matplotlib
python run_experiments.py --task sweep      --seeds 15 --T 4000 \
    --out results.json
python run_experiments.py --task ablation --seeds 8  --T 4000 \
    --out ablation.json
python run_experiments.py --task scale     --seeds 5  --T 4000 \
    --sizes 7x7,11x11,15x15 --out scaling.json
python make_figures.py
python test_biomaterial_net.py
```

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